

SOFTWARE REQUIREMENTS SPECIFICATION

for

HOSPITAL MANAGEMENT SYSTEM (HMS)

Version 1.0 Approved

Prepared by:

Anish Chitnis (24CE1044)

Aaryan Choube (24CE1045)

Somansh Dafade (24CE1047)

Batch: A / A1

Department of Computer Engineering

Ramrao Adik Institute of Technology, DY Patil University, Navi Mumbai

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a detailed description of the Hospital Management System (HMS). This document defines the functional and non-functional requirements, operating constraints, external interfaces, and expected behavior of the system. It serves as a baseline for system design, development, testing, and validation, and establishes a common understanding among patients, doctors, nurses, receptionists, pharmacists, laboratory staff, hospital administrators, developers, and testers.

1.2 Document Conventions

This document follows the structure and conventions of the IEEE-style SRS template prescribed for the Software Engineering laboratory. Section numbering follows a hierarchical format. The word "shall" indicates a mandatory requirement, while "should" indicates a recommended requirement. Functional requirements are uniquely identified using the format FR-x.y, where x represents the system feature and y represents the requirement number. Requirement priorities are classified as High, Medium, or Low. As this project follows the V-Model, every functional requirement documented in Section 4 is intended to map directly to a corresponding verification and validation (test) activity performed during the testing phases of the project.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers, project guides or mentors, testers, hospital administrators, doctors, and other stakeholders involved in the development and use of HMS. Readers should first review Section 1 for the purpose and scope of the system, followed by Section 2 for the overall system description. Section 3 describes external interfaces, Section 4 contains detailed system features and functional requirements, and Section 5 defines non-functional requirements and quality constraints.

1.4 Product Scope

The Hospital Management System is a web-based application designed to automate and streamline hospital administration activities that are otherwise handled manually or through fragmented registers and spreadsheets. The system manages patient registration, appointment scheduling, doctor and department management, billing and payments, pharmacy inventory, laboratory test and diagnostic report management, and notices. The system aims to reduce paperwork, minimize scheduling and record-keeping errors, improve transparency, and provide faster access to healthcare services for patients and hospital staff.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Roger S. Pressman, Software Engineering: A Practitioner's Approach, McGraw-Hill.

- Institutional hospital rules and policies applicable to registration, billing, pharmacy, and laboratory operations.

2. Overall Description

2.1 Product Perspective

HMS is a new, self-contained, web-based client-server application connected to a centralized database. It replaces manual and register-based hospital administration processes with a centralized digital system. Users access the system through modern web browsers on computers and mobile devices. The project is being developed following the V-Model of the software development life cycle, in which each phase of development (requirements, system design, architecture design, module design, and coding) has a directly corresponding phase of testing (acceptance testing, system testing, integration testing, and unit testing respectively). This ensures that every requirement captured in this SRS is validated against a specific testing activity before the system is deployed. The system may optionally integrate with an external insurance or government health-scheme portal in future versions.

2.2 Product Functions

- Patient registration and appointment scheduling.
- Doctor, department, and ward/bed management.
- Billing and payment management with digital receipts.
- Pharmacy and medicine inventory management.
- Laboratory test request and diagnostic report management.
- Notice and announcement broadcasting.
- Administrative reports for hospital management and decision-making.

2.3 User Classes and Characteristics

User Class	Description / Access
Hospital Administrator	Has full access to manage staff, departments, fee structures, inventory, and hospital-wide reports.
Doctor	Views assigned patient records, records diagnoses and prescriptions, and updates treatment status.
Nurse / Ward Staff	Records vitals, updates bed/ward status, and assists in patient care scheduling.
Receptionist / Front Desk	Registers patients, schedules appointments, and manages OPD queue and admissions.
Patient	Books appointments, views prescriptions and reports, and pays bills online.

User Class	Description / Access
Pharmacist	Manages medicine stock, dispenses medicines against prescriptions, and updates inventory.
Laboratory Technician	Manages test requests, uploads diagnostic reports, and updates test status.

Role-based access control shall be used to ensure that each user class can access only the functions and data relevant to its role.

2.4 Operating Environment

The system shall operate on a server environment supporting a web application backend and a relational database. The client application shall be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. The system shall support Windows and Linux server environments. The client requires an internet-connected desktop, laptop, tablet, or smartphone with a supported web browser.

2.5 Design and Implementation Constraints

- The system shall implement secure authentication and password hashing.
- Role-based access control shall be enforced for all protected modules.
- The system shall use a relational database such as MySQL or PostgreSQL for persistent data storage.
- The system shall use secure HTTPS communication for client-server communication.
- The system shall comply with institutional and medical data-privacy policies regarding patient records.
- Each module developed shall have a corresponding test plan defined at the requirement stage, consistent with the V-Model.

2.6 User Documentation

The system shall be accompanied by user documentation and short help guides for the Administrator, Doctor, Nurse, Receptionist, Pharmacist, Laboratory Technician, and Patient user classes. The documentation shall provide basic login instructions, feature usage steps, and basic troubleshooting guidance.

2.7 Assumptions and Dependencies

It is assumed that users are provided valid login credentials by the hospital administration and possess basic computer or smartphone literacy. The system depends on reliable internet connectivity and the availability of a database server. Online bill payments depend on the availability of a secure third-party payment gateway. The system may depend on an external insurance or government health-scheme portal in future versions for verifying patient eligibility.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide role-specific login and dashboard interfaces. The Administrator dashboard shall provide hospital-wide management functions. The Doctor dashboard shall provide patient records, prescription entry, and appointment functions. The Patient dashboard shall provide appointment booking, bill payment, and report-viewing functions. Nurses, Receptionists, Pharmacists, and Laboratory Technicians shall each receive dashboards relevant to their assigned duties. Interfaces shall be responsive and usable on desktop and mobile browsers.

3.2 Hardware Interfaces

The system shall support access from internet-enabled desktops, laptops, tablets, and smartphones. A printer may optionally be connected at the reception and pharmacy counters for printing receipts and prescriptions. Biometric attendance or patient-identification hardware may be integrated in a future version.

3.3 Software Interfaces

The system shall interface with a relational database management system such as MySQL or PostgreSQL for persistent data storage. It shall interface with a third-party payment gateway API for online bill payments. The application shall communicate with these software components using secure protocols and standard APIs. Future versions may interface with an external laboratory information system or insurance portal.

3.4 Communications Interfaces

Communication between the client browser and the application server shall use HTTPS. The system shall use standard HTTP/HTTPS-based communication for web requests and API communication. Email or SMS notification services may be used for appointment reminders, report availability, and bill-due reminders. Authentication and sensitive data transfer shall use secure communication mechanisms.

4. System Features

4.1 Patient Registration and Appointment Management

4.1.1 Description and Priority

Allows the Receptionist to register patients and schedule appointments with available doctors. Priority: High.

4.1.2 Stimulus/Response Sequences

- Receptionist enters patient details and submits the form → the system validates the input, generates a unique Patient ID, and creates the patient record.
- Receptionist selects a doctor, date, and time slot and confirms booking → the system checks doctor availability and, if free, books the appointment and updates the schedule.

- Receptionist attempts to book an already occupied time slot → the system rejects the booking and displays an appropriate error message.

4.1.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-1.1	The system shall allow the Receptionist/Admin to add, modify, delete, and view patient records and generate a unique Patient ID for each patient.	High
FR-1.2	The system shall display doctor availability by department, date, and time and support appointment booking based on the selected slot.	High
FR-1.3	The system shall prevent double booking of the same doctor time slot and maintain an accurate appointment schedule.	High
FR-1.4	The system shall reject invalid or incomplete patient information and display an appropriate validation message.	Medium

4.2 Billing and Payment Management

4.2.1 Description and Priority

Manages consultation, treatment, and pharmacy bill calculation and payment collection. Priority: High.

4.2.2 Stimulus/Response Sequences

- A consultation, treatment, or pharmacy transaction is completed → the system computes the applicable bill amount.
- The Patient initiates a payment → the system sends the transaction to the payment gateway and, on success, generates a digital receipt.
- A bill due date approaches → the system displays or sends a due-date reminder.

4.2.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-2.1	The system shall calculate consultation, treatment, and pharmacy charges based on applicable hospital tariffs.	High
FR-2.2	The system shall record online and offline payment transactions and generate digital receipts.	High
FR-2.3	The system shall track pending dues for each patient and notify patients of upcoming due dates.	Medium
FR-2.4	The system shall display an appropriate message when a payment transaction fails or is cancelled.	High

4.3 Doctor and Department Management

4.3.1 Description and Priority

Enables the Administrator to manage doctors, departments, and ward/bed availability. Priority: Medium.

4.3.2 Stimulus/Response Sequences

- Admin adds or updates a doctor's profile and department assignment → the system saves the details and updates the doctor directory.
- Admin updates bed/ward occupancy status → the system reflects the updated availability across the hospital dashboard.
- Admin or Doctor requests a schedule report → the system generates the doctor-wise or department-wise report for the selected period.

4.3.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-3.1	The system shall allow the Admin to add, modify, and view doctor profiles along with their assigned department and consultation timings.	Medium
FR-3.2	The system shall track ward and bed occupancy status in real time.	Medium
FR-3.3	The system shall generate department-wise and doctor-wise activity reports.	Medium

4.4 Pharmacy and Inventory Management

4.4.1 Description and Priority

Allows the Pharmacist to manage medicine stock and dispense medicines against prescriptions. Priority: High.

4.4.2 Stimulus/Response Sequences

- Doctor submits a prescription → the system forwards it to the Pharmacy module for dispensing.
- Pharmacist dispenses medicines against a prescription → the system deducts the dispensed quantity from stock and updates the pharmacy bill.
- Medicine stock falls below the reorder threshold → the system generates a low-stock alert.

4.4.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-4.1	The system shall allow Doctors to create digital prescriptions linked to a patient's appointment record.	High
FR-4.2	The system shall allow the Pharmacist to dispense medicines against a valid prescription and update stock quantities.	High
FR-4.3	The system shall generate a low-stock alert when medicine quantity falls below a predefined threshold.	Medium
FR-4.4	The system shall prevent dispensing of expired or out-of-stock medicines.	Medium

4.5 Laboratory and Diagnostic Report Management

4.5.1 Description and Priority

Allows Laboratory Technicians to manage diagnostic test requests and reports. Priority: Medium.

4.5.2 Stimulus/Response Sequences

- Doctor requests a diagnostic test for a patient → the system creates a test request with status Pending.
- Laboratory Technician uploads the test result → the system updates the status to Completed and notifies the Doctor and Patient.
- Patient views a completed report → the system displays the report in the Patient dashboard.

4.5.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-5.1	The system shall allow Doctors to raise diagnostic test requests for a patient.	Medium
FR-5.2	The system shall allow Laboratory Technicians to update test status and upload diagnostic reports.	Medium
FR-5.3	The system shall notify the Doctor and Patient when a diagnostic report becomes available.	Medium
FR-5.4	The system shall maintain a history log of all diagnostic tests conducted for a patient.	Low

4.6 Notice and Report Management

4.6.1 Description and Priority

Allows the Administrator to broadcast notices and generate hospital-wide administrative reports.

Priority: Low.

4.6.2 Stimulus/Response Sequences

- Admin composes a notice and selects the target audience → the system publishes the notice to the relevant dashboards.
- Admin requests an administrative report (patient inflow, revenue, occupancy) → the system generates the report for the selected period.

4.6.3 Functional Requirements

Req ID	Requirement Description	Priority
FR-6.1	The system shall allow the Admin to post notices visible to all or selected staff and patients.	Low

Req ID	Requirement Description	Priority
FR-6.2	The system shall allow the Admin to generate patient-inflow, revenue, and occupancy reports.	Low

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall support at least 300 concurrent users, appropriate for a mid-sized hospital handling several hundred patients daily, under normal operating conditions. The average response time for standard operations such as viewing dashboards, submitting forms, and retrieving records should be less than 3 seconds under normal network and server conditions. Report generation should complete within a reasonable time based on the volume of data requested.

5.2 Safety Requirements

The system shall protect sensitive patient information and prevent accidental loss of important medical records. Regular database backups shall be maintained. The system shall provide safeguards against accidental deletion or unauthorized modification of patient, billing, prescription, and diagnostic records.

5.3 Security Requirements

The system shall authenticate users before granting access to protected functions. Passwords shall be stored using secure password hashing. Role-based access control shall be enforced for all modules. Sensitive communication shall use HTTPS. Payment transactions shall be processed through a secure third-party payment gateway, and the system shall not directly store sensitive card information. Access to personal patient health information shall be restricted according to user roles, in line with applicable data-privacy norms.

5.4 Software Quality Attributes

The system shall be reliable, maintainable, scalable, and user-friendly. The system shall use modular components to support future enhancement and to allow independent unit and integration testing consistent with the V-Model. Error logging shall be implemented to assist debugging and maintenance. The interface shall be simple enough for users with basic computer or smartphone literacy.

5.5 Business Rules

Appointment scheduling, billing, pharmacy dispensing, and diagnostic reporting shall follow the hospital policies defined by the administration. Exceptions such as fee waivers, emergency admissions, or priority test processing shall require appropriate authorization and shall be recorded in the system for accountability.

6. Conclusion

This SRS document establishes a clear and common understanding between the project team and all stakeholders regarding what the Hospital Management System is expected to do. By clearly defining functional requirements (FR-x.y), non-functional requirements, external interfaces, and user classes, this document reduces ambiguity during design and development and provides a baseline for estimating effort and schedule. Since the project follows the V-Model, every requirement captured here also serves as the basis for a corresponding verification/validation activity, ensuring that the final system can be objectively tested and validated against what has been specified in this document.